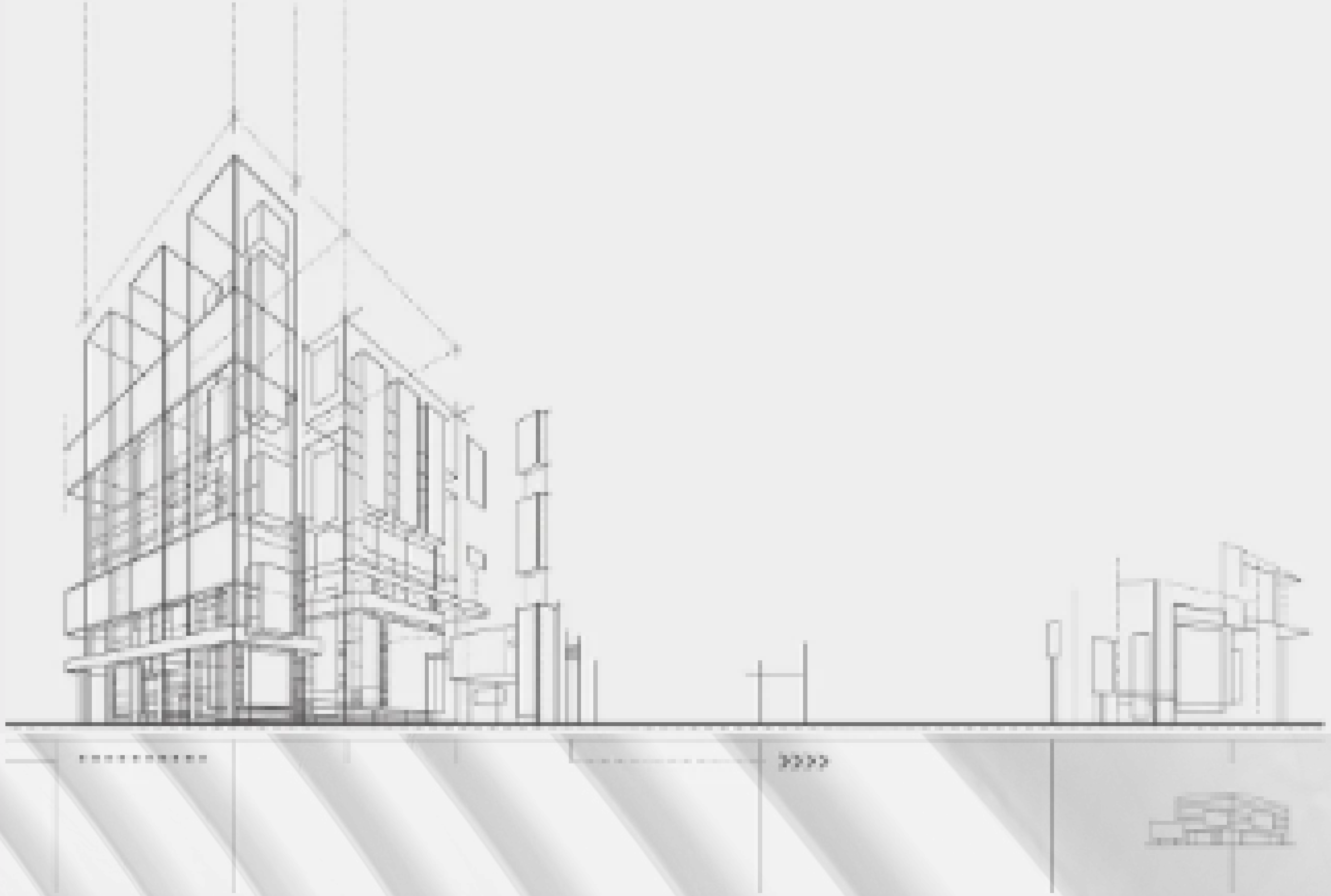
BIM PORTFOLIO



Maryam Shojaei
BIM Professional

2020-2025

ABOUT ME

Works

- **VDC & BIM Management:** Led clash detection for 30+ story high-rise, resolving 20+ interdisciplinary conflicts.
- **Digital Fabrication:** Coordinated CNC operations for 300,000 Sq. Ft ACM panel project, optimizing material usage.
- **Structural Design:** Produced fabrication drawings and 3D models for 50m+ telecom towers.

Education

- Postgraduate Certificate, BIM Management (George Brown College)
- Master of Architecture
- Certifications: Working at Heights, Health & Safety Awareness

Achievements

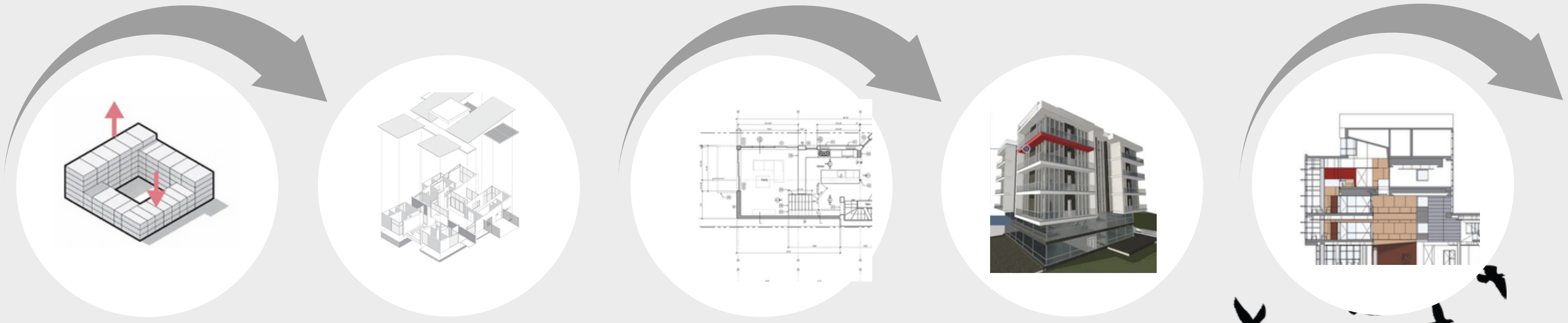
- Reduced design revisions by 15% through system optimization and precise material take-offs.
- Improved project delivery timelines by 10% by streamlining fabrication coordination.
- Achieved 100% material usage on a 300,000 Sq. Ft facade project.
- Streamlined design verification by 20% through accurate model preparation.

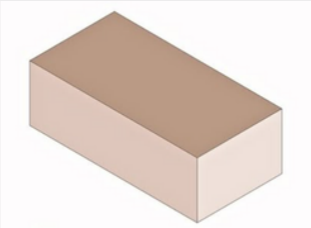
Skills

- **Software:** Revit, Navisworks, ACC/BIM 360, Civil 3D, AutoCAD, SolidWorks, FARO/LIDAR
- **BIM & Technical:** VDC Coordination, BIM Execution Planning (ISO 19650), Clash Detection, Reality Capture

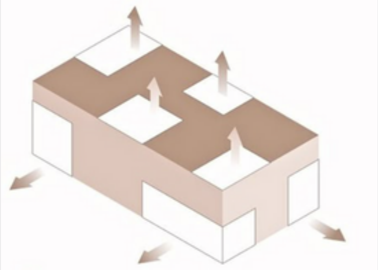
Associate Creative Director

- Model Delivery: IFC + native formats, milestone-based submissions
- Coordinates: Shared base point & consistent coordinate system
- Issue Management: BCF workflow for tracking and resolving design issues
- Meetings: BIM kickoff, coordination meetings, monthly progress reviews
- Quality Control: Visual, clash, standards, and integrity checks

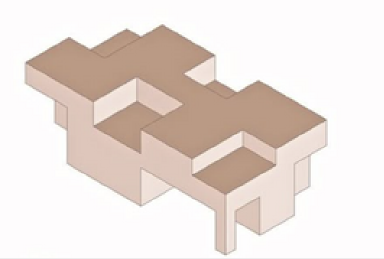




Build Mass



Program Division



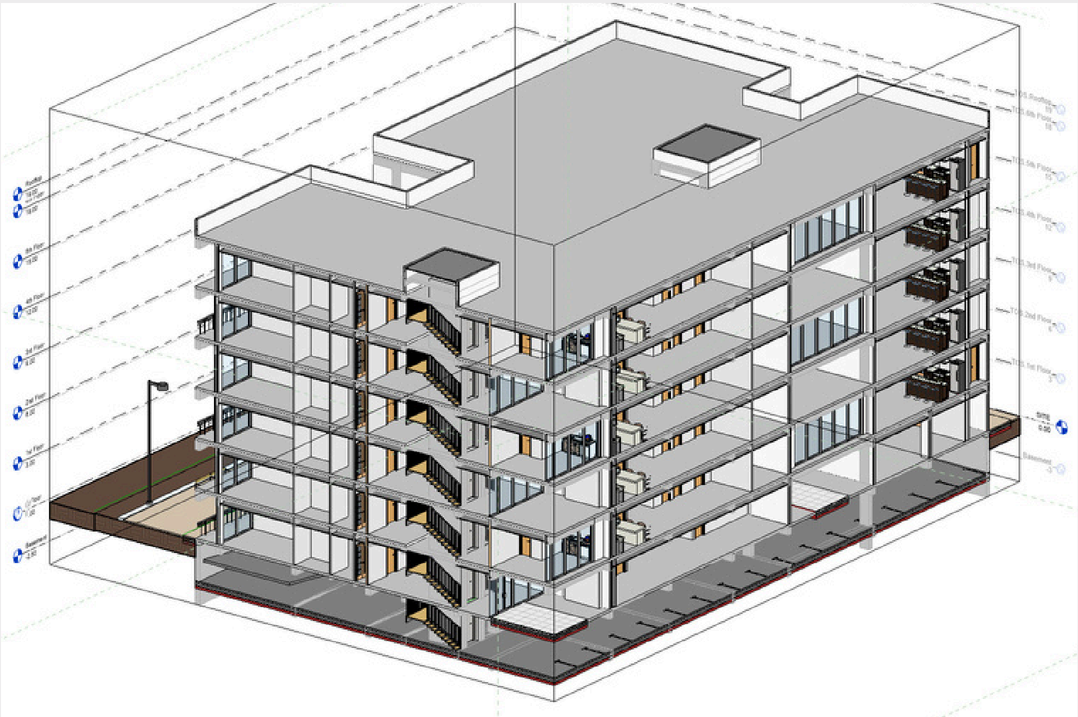
Mass Subtraction

Final form evolution

Managed a multi-disciplinary BIM project using Autodesk Construction Cloud (ACC) as the centralized Common Data Environment (CDE). The platform facilitated real-time collaboration and model coordination between architecture, structure, and MEP teams, ensuring a single source of truth for all project data.

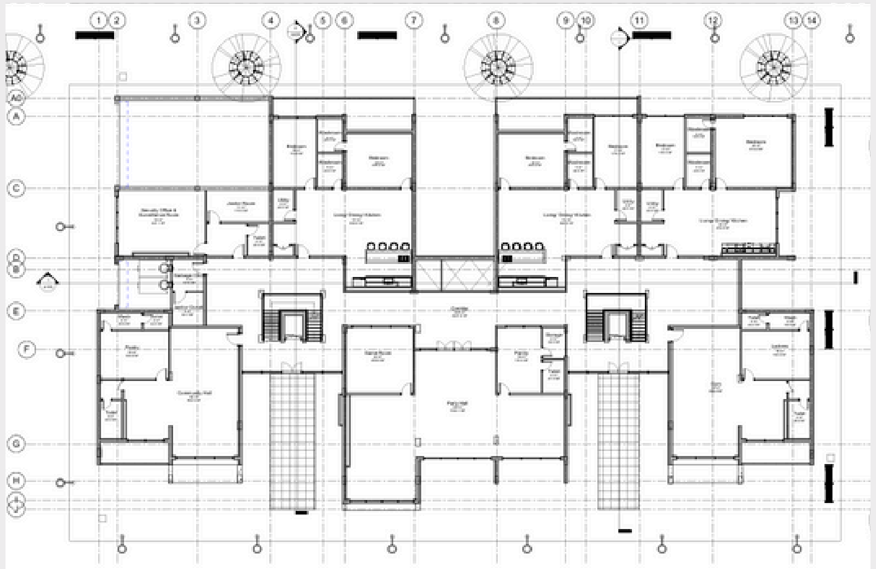
Utilized ACC's clash detection and real-time platform to resolve design conflicts early and ensure all stakeholders worked from a single source of truth.

Category	Detail
Project	39 Raglan Ave, York Ward, Toronto
Official Plan	Apartment Neighbourhoods
Primary Zone	R(d0.6)
Permitted Density	FSI 0.6
Max Height (Primary)	12 m
Max Height (RA/RM2)	24 m (~8 storeys)
Setbacks	Front: 6.0 m Rear: 7.5 m Sides: 7.5 m



First Floor Plan

PROJECT SCOPE

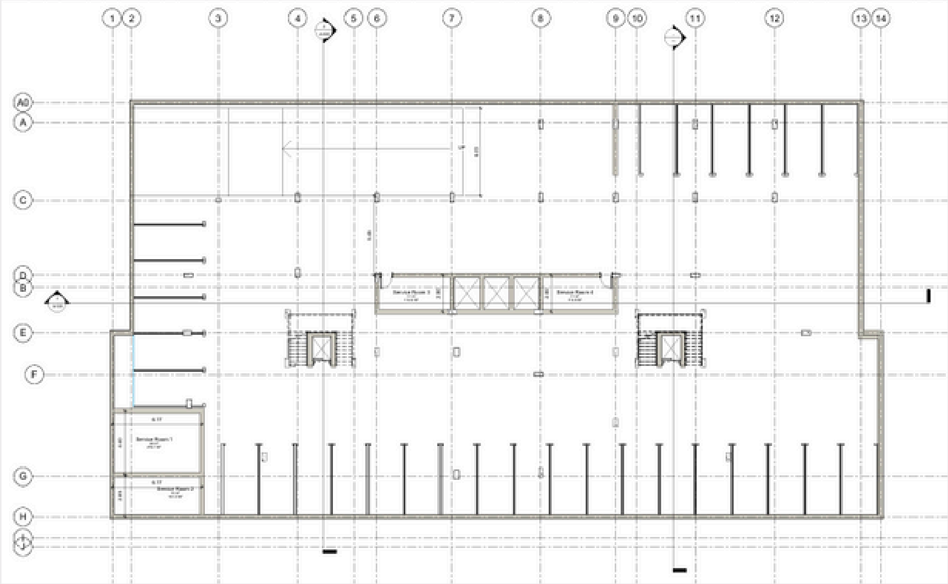


Ground Floor Plan

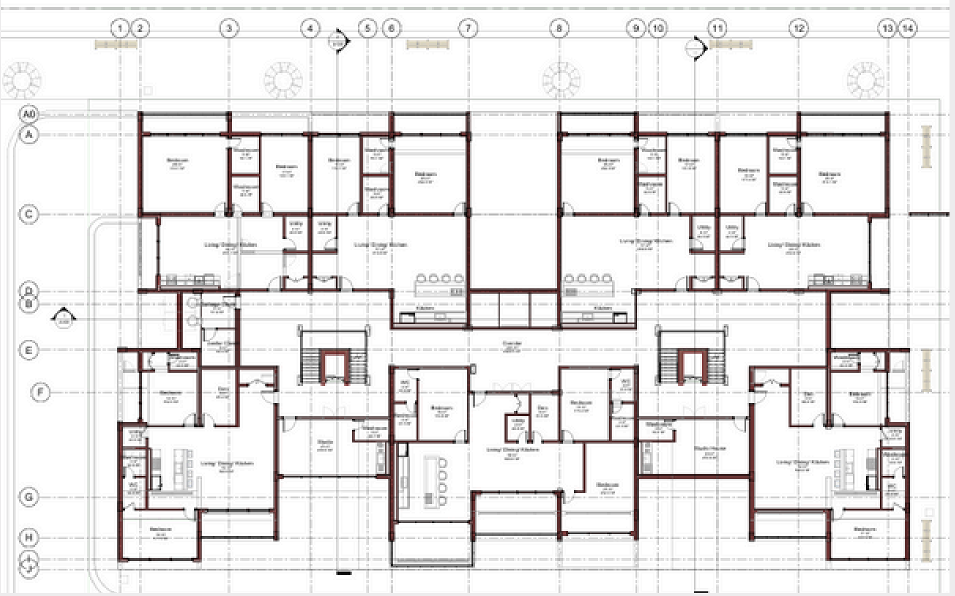
Location: Toronto, ON
Project Type: 4-storey to 6-storey vertical expansion with commercial (ground floor) and residential units, plus underground parking.

Key Deliverables:

- Architectural & structural redesign for 2 additional floors
- Full MEP systems integration
- OBC-compliant fire protection & accessibility systems
- LOD 300+ BIM models & construction documents



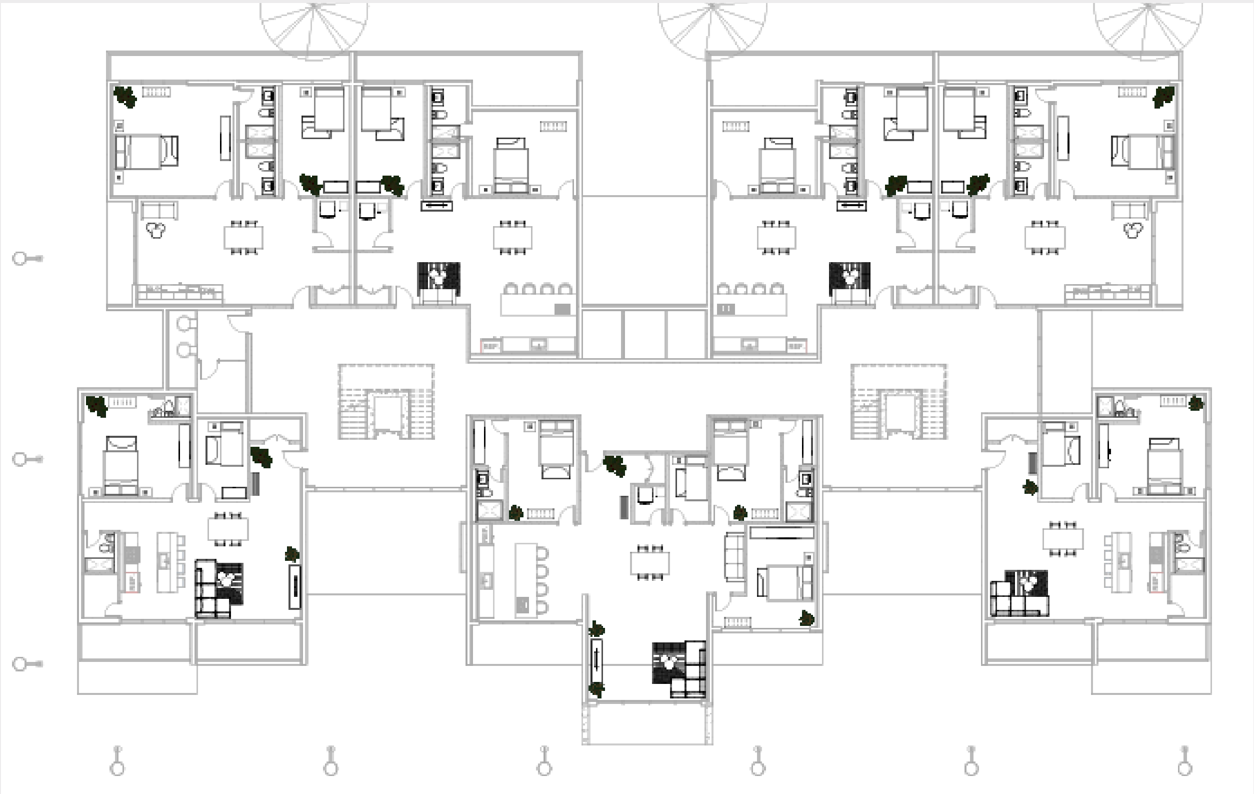
Basement



First Floor Plan

Key Achievements

- Added 12 units during active commercial operation
- Integrated new and existing building systems
- Secured full permitting
- Delivered construction-ready BIM models
- Demonstrates expertise in urban vertical expansion and mixed-use projects.



First Floor Plan

We used Plannerly to develop and manage the BIM Execution Plan (BEP) for this project. It provided a structured and collaborative environment to define project goals, team responsibilities, workflows, and BIM uses.

INTERIOR DESING

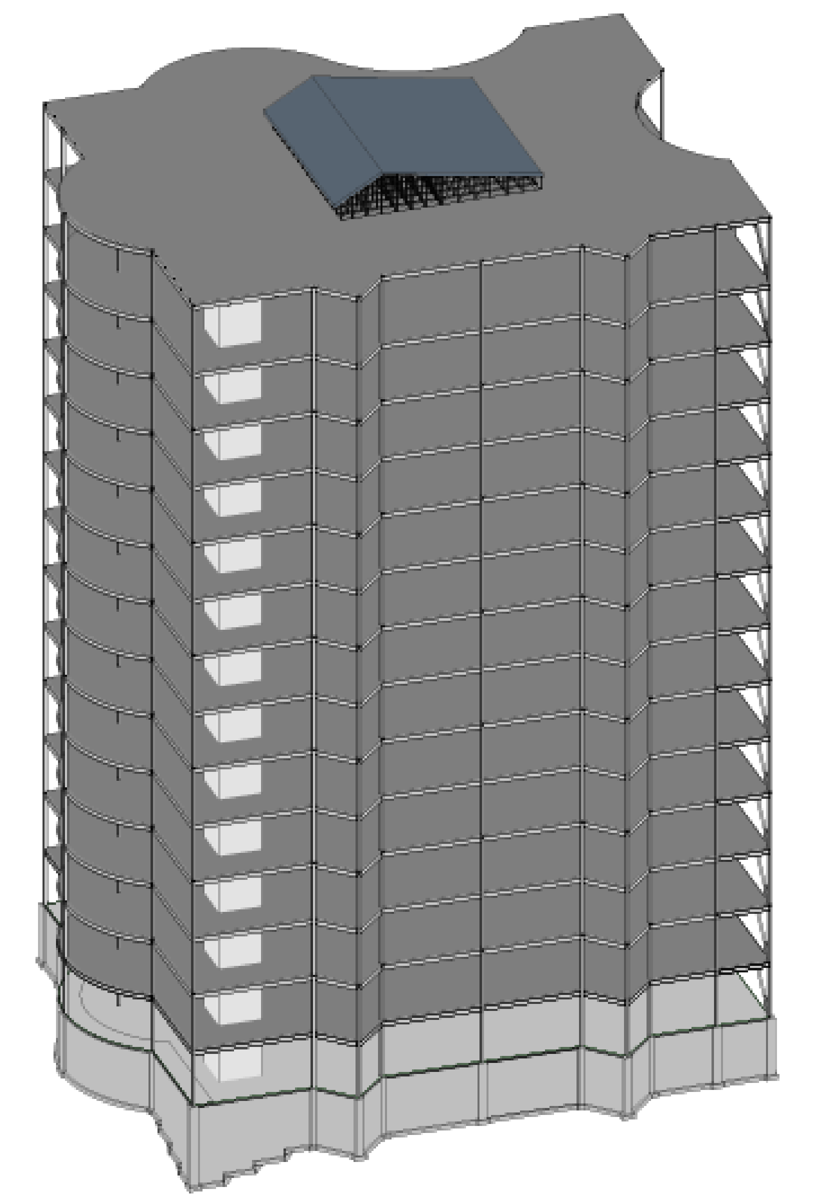
Furniture Schedule

Furniture Schedule			
Image	Count	Family and Type	Level
	8	Artificial banana plant 1500mm: Ny märkning	First Floor
	29	Artificial ZZ plant 1100mm: Ny märkning	First Floor
	13	Bed_Wood_Frame_And_Headboard_Classic_502_BIMLib: Qween Size	First Floor
	28	Bedside Table - Nightstand: Bedside Table - Nightstand	First Floor
	7	Cabinet QBUS 4 shelves base frame handle 2020x400x420 mm: Cabinet QBUS 4 shelves base frame handle 2020x400x420 mm	First Floor
	5	Furniture_Bed_2: 1422x1981x330mm-Double	First Floor
	4	Furniture_Chair_Dining: 385x320x425mm	First Floor
	24	Furniture_Other-Furniture_Fritz-Hansen_Rug-balance: Default	First Floor
	3	Furniture_Other-Furniture_Fritz-Hansen_Rug-Dots: Default	First Floor
	3	Furniture_Shelving&Storage_Essem-Design_Tamburin_Shoe-Rack: Tamburin shoe rack - 1000	First Floor
	3	Furniture_Sofas-Armchairs_Icons-of-Denmark_ EC1-Sofa_Configuration2: Default	First Floor

Furniture Schedule			
Image	Count	Family and Type	Level
	7	Furniture_Tables_CIDER-LA-MANUFACTURE_Trio-Coffee-Table: Default Structure - (Black chrome) Plateau - (Noce Canaletto)	First Floor
	1	Furniture_Tables_Plank_Mart-Table-Round-h-73: Mart Table Round-Mod. 9835-01 (Ø 120cm) - Top HPL Black - Structure Black Aluminum	First Floor
	2	Furniture_Wardrobe: 1500x1800x600mm	First Floor
	1	Furniture_Wardrobe: 1500x1800x600mm 2	First Floor
	4	INSIDE MESA CENTRO 120X45: INSIDE MESA CENTRO 120X45	First Floor
	3	Media_Center_Console_Parametric_740_BIMLibrary.co: Media_Center_Console_Parametric_740_BIMLibrary.co	First Floor
	11	Ottoman_Classic_Storage_765_BIMLib: 4'-0"W X 1'-6"D X 1'-6"H	First Floor
	6	Sofa_Corbu: Sofa	First Floor
	2	SSG_Firehouse_Bed_3_Drawers_LAM: Default	First Floor
	1	study_table_900x450: Default	First Floor
	7	Table_Dining_Chairs_Simple: Table w 4 Seats	First Floor
	7	Table_Simple_Rectangular: 1830 x 0762mm	First Floor

Plannerly allowed all stakeholders to agree on clear deliverables from the beginning, helping us align expectations and ensure consistent project execution.

STRUCTURAL BIM PROJECT

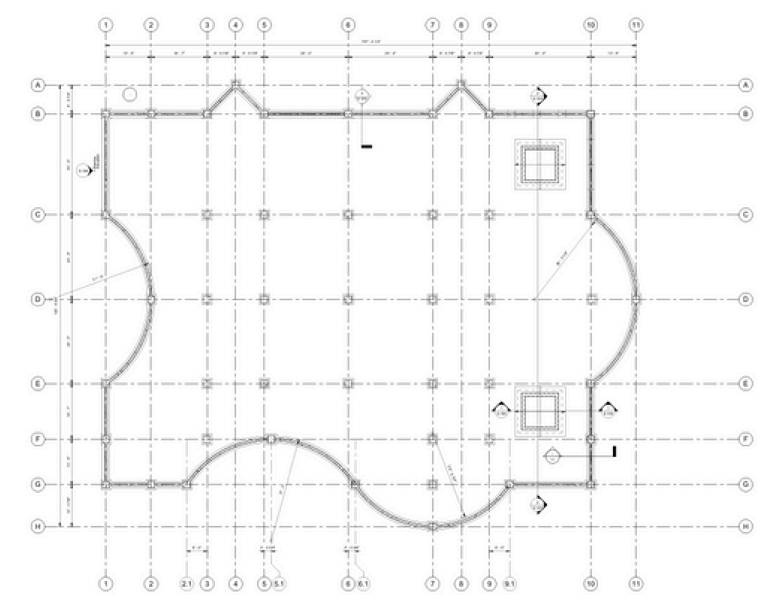


Isometric view

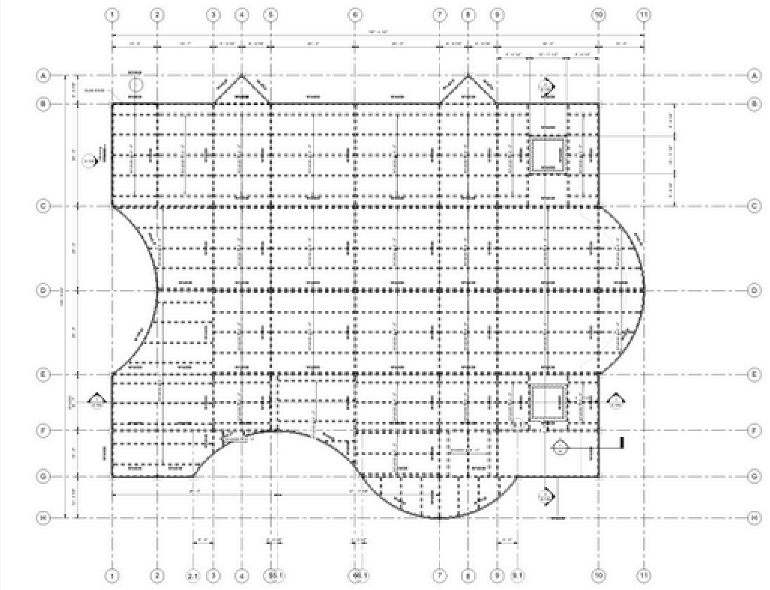
I specialize in developing detailed, constructible Structural BIM models in Autodesk Revit, with demonstrated expertise in:

- **Structural Framing:** Establishing project grids, columns, and vertical load paths
- **System Detailing:** Modeling foundations, beams, slabs, and structural connections
- **Reinforcement Design:** Detailing rebar placement for concrete elements to ensure structural integrity

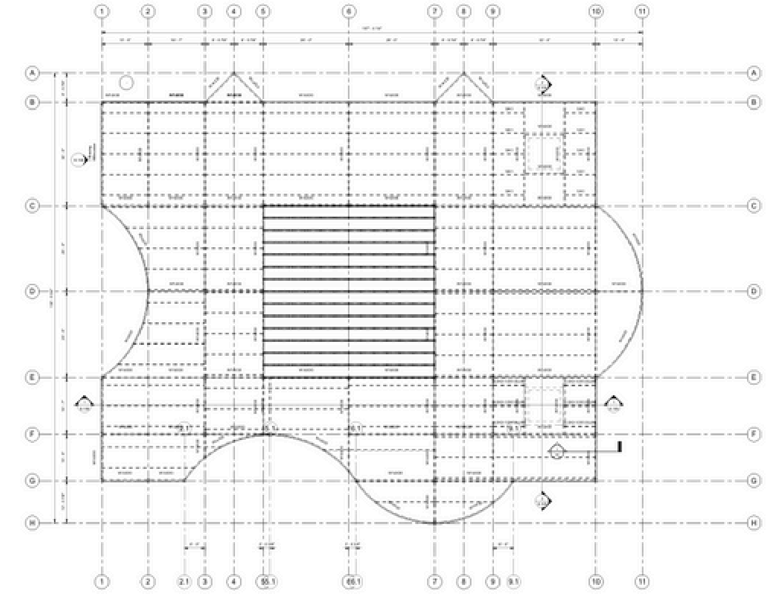
My approach integrates precise modeling with practical constructibility, ensuring coordinated, fabrication-ready deliverables.



Footing



Floor Plan



Roof

PROJECT PHASES & KEY DELIVERABLES

Phase 1: Project Initiation

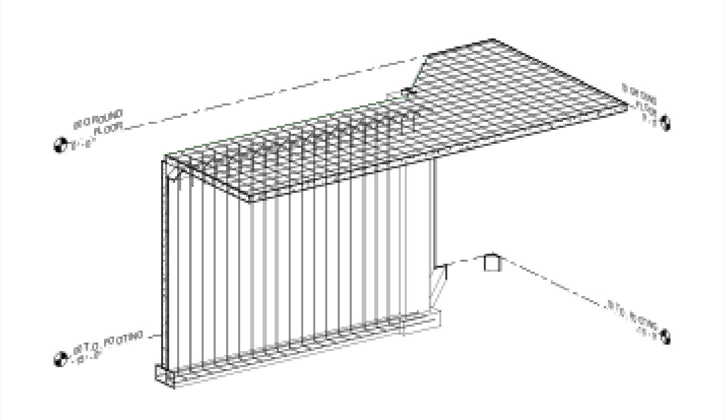
- **Configured Revit templates and standards for high-rise structural design**

Phase 2: Primary Structural Layout

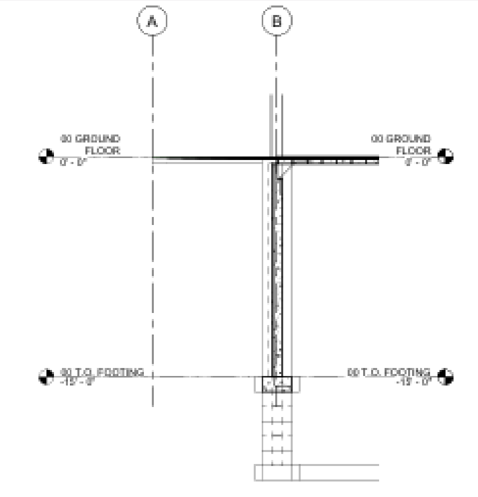
- **Established structural grid and placed vertical column system**

Phase 3: Foundation Systems

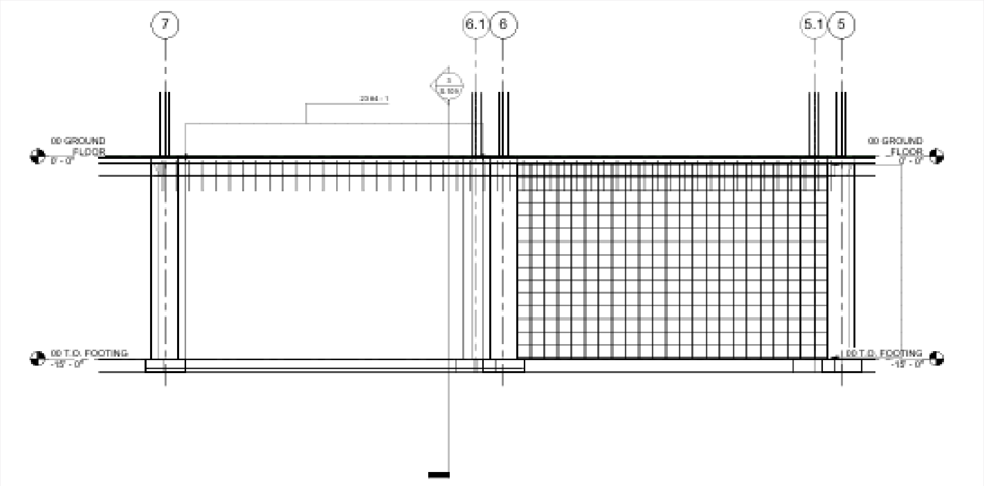
- **Modeled high-capacity substructure including core walls and pile caps**



3D Foundation Wall



3D Foundation Wall Section



Foundation Rebar Elevation

Phase 4: Superstructure Framing

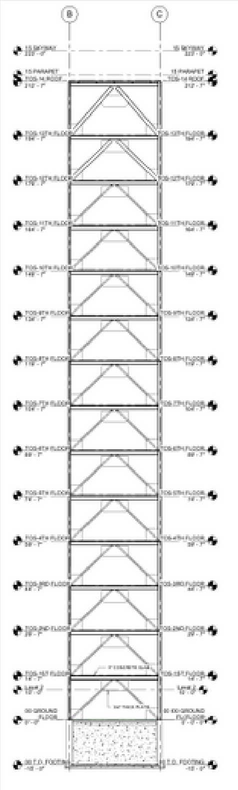
Developed the horizontal framing and lateral systems with beams, bracing, and shear walls for stability.

Phase 5: Slab Design & Coordination

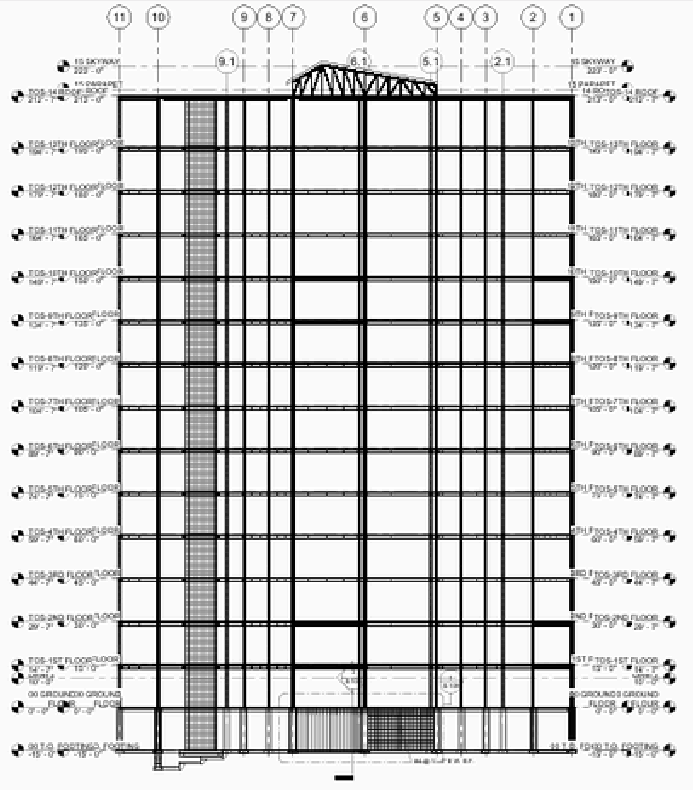
Created composite structural slabs for all 15 levels, integrating coordinated shaft openings for MEP systems.

Phase 6: Detailed Reinforcement

Added detailed rebar to all concrete elements (columns, walls, slabs), preparing the model for fabrication and construction.

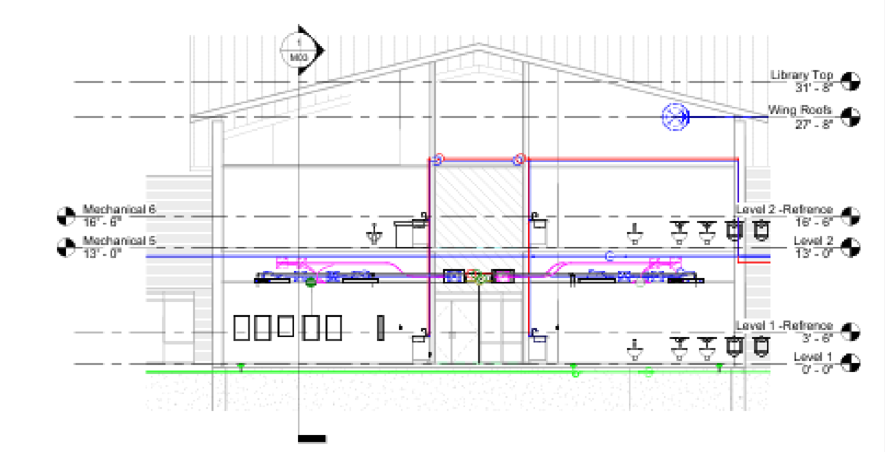


Bracing Elevation

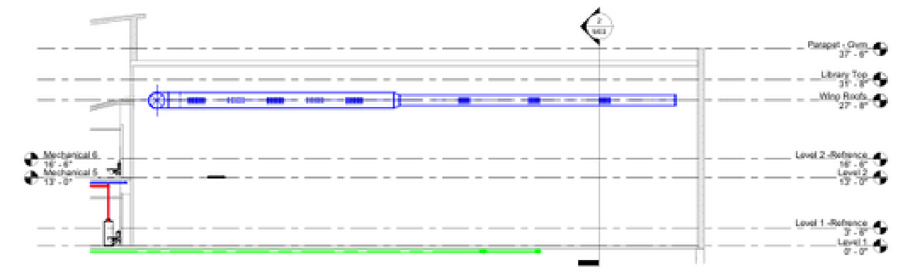


North Section

MECHANICAL BIM SYSTEMS

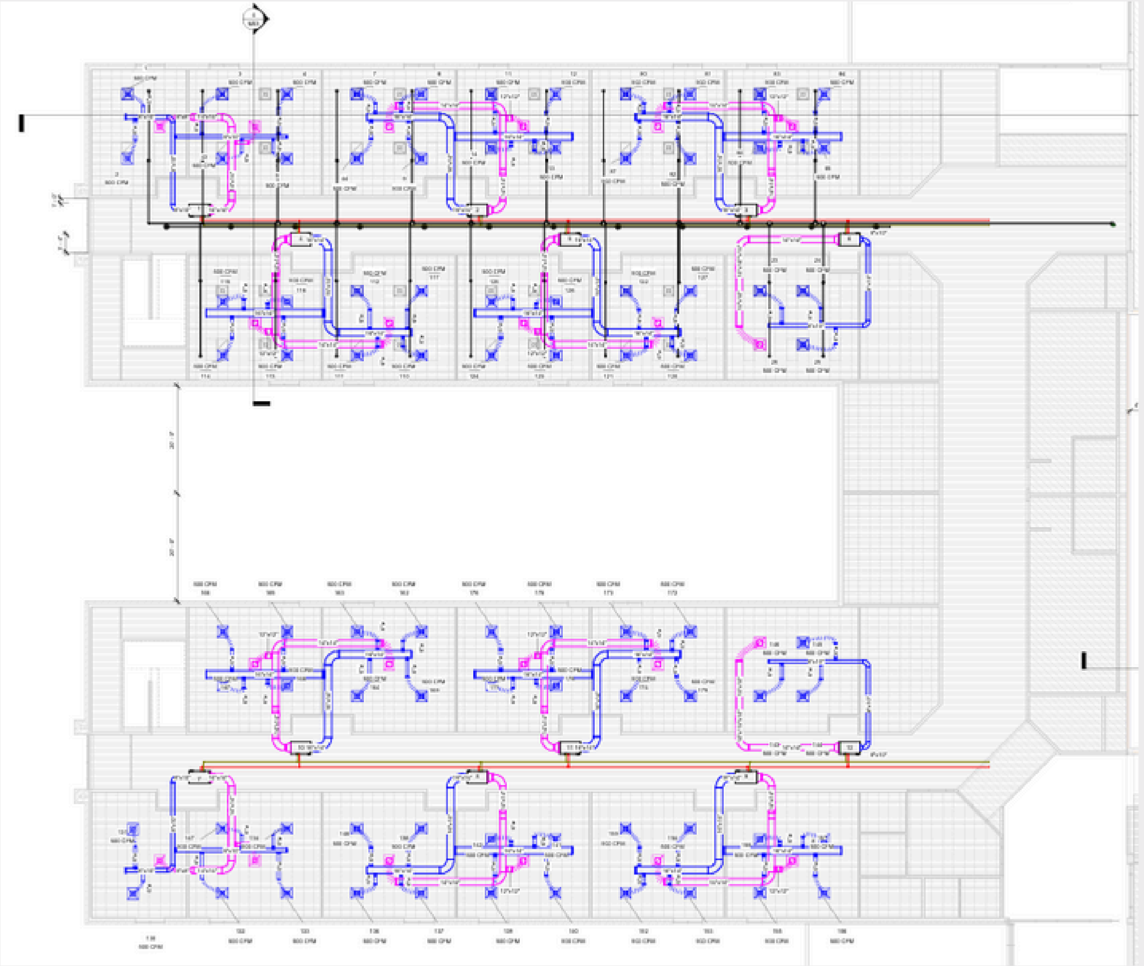


Section

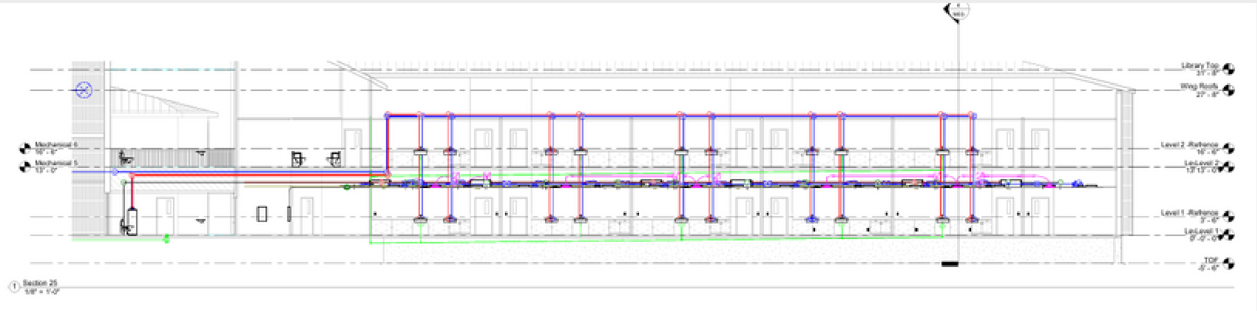


Section

Mechanical Design: This project features a complete, BIM-driven mechanical design for a two-level educational building. The Revit model integrated a high-efficiency Variable Refrigerant Flow (VRF) system, detailed ductwork, and plumbing layouts. Central to the design was ensuring optimal indoor air quality and thermal comfort for classroom environments. The model was crucial for spatial coordination, enabling seamless clash detection with structural and electrical systems to preempt on-site conflicts and streamline installation.

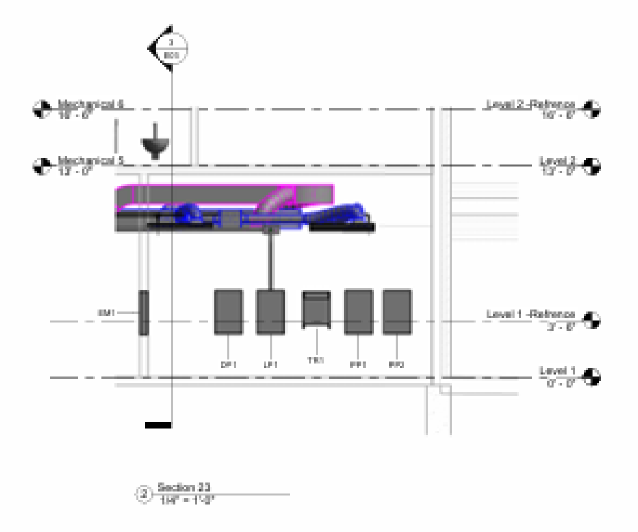


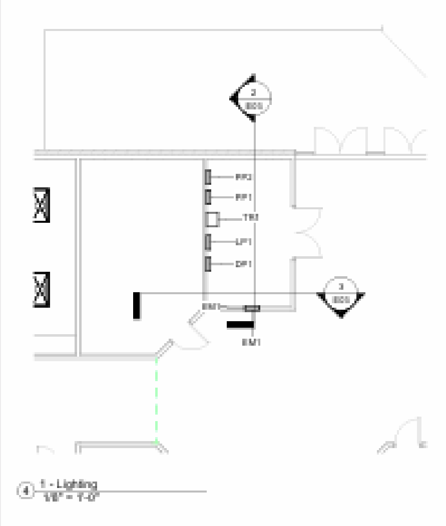
Ceiling Mech

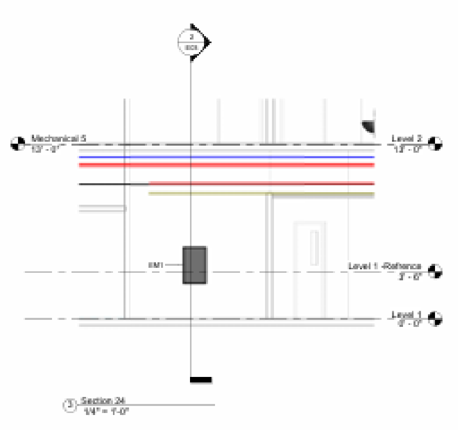


ELECTRICAL IN BIM SYSTEMS

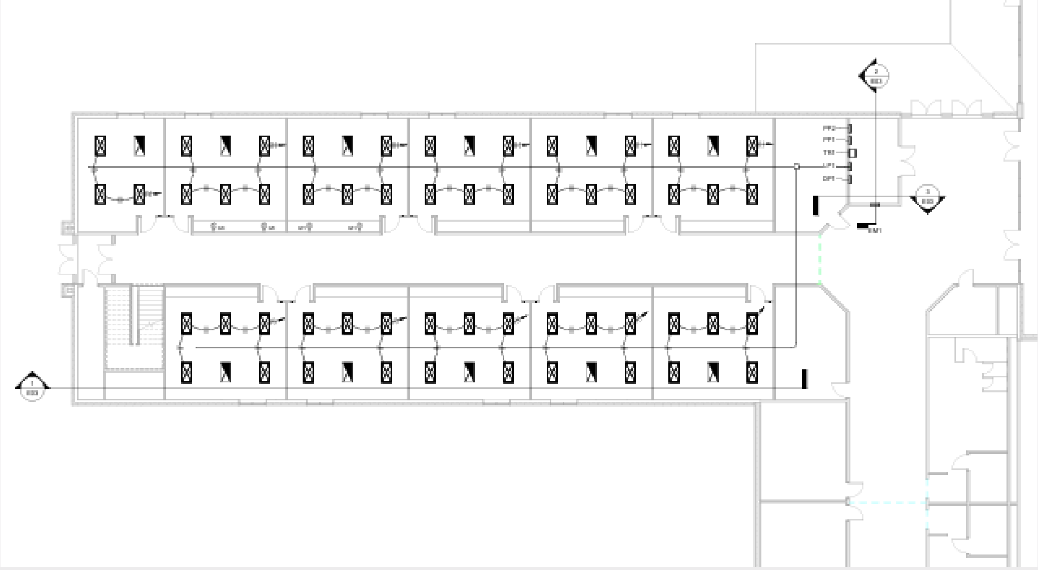
- **Voltage Level:** 120/208V
- **Frequency:** 60 Hz
- **Outlets Mounting Height** - 300 mm
- Switches Mounting Height -**1200 mm**
- **Cable Tray:** 300x100 mm Lighting
- **Switch:** Single-pole
- **Code Compliance:** Ontario Electrical Safety Code 2021





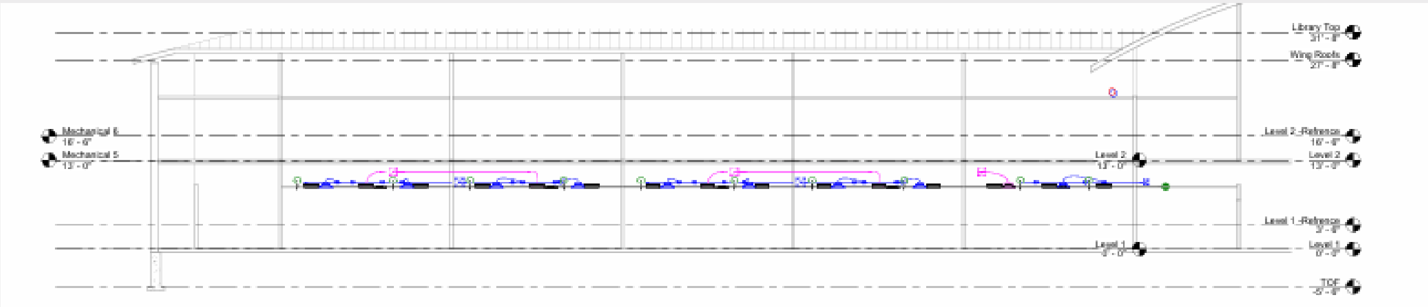


This project showcases a fully coordinated BIM-based electrical design for a two-level educational facility. Developed in Revit, the model integrates power, lighting, and low-voltage systems within a federated BIM environment.



Power Plan

This approach enabled proactive clash detection with structural and mechanical systems and facilitated the production of construction-ready drawings, ensuring a efficient and code-compliant installation.

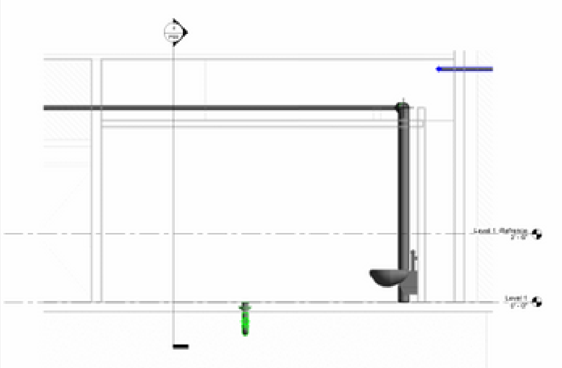


Section

PLUMBING BIM SYSTEMS

I specialize in delivering fully compliant mechanical and plumbing systems through rigorous coordination and quality assurance. My expertise includes:

- Regulatory Management: Preparing and submitting technical drawings to AHJs and Fire Departments for approval
- Integrated Coordination: Leading 3D spatial coordination between MEP, structural, and architectural systems
- Technical Detailing: Specifying critical components including isolation valves, accessible shut-offs, and water hammer arrestors
- Quality Control: Ensuring all systems meet code requirements and manufacturer specifications



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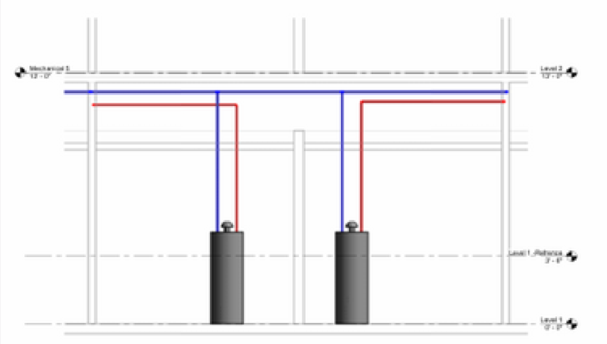
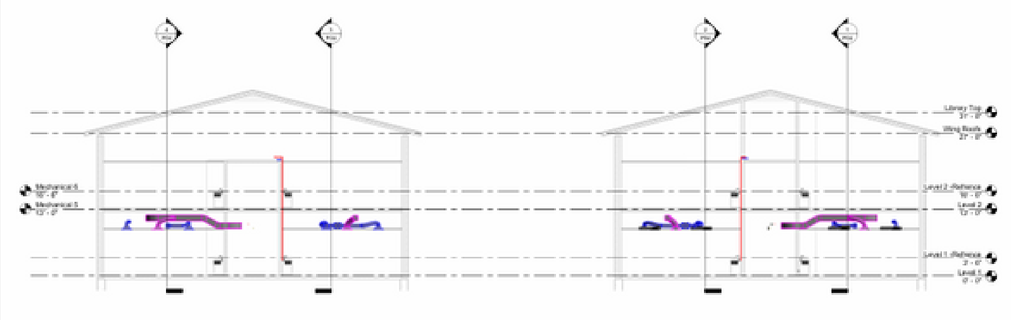


Mechanical Room



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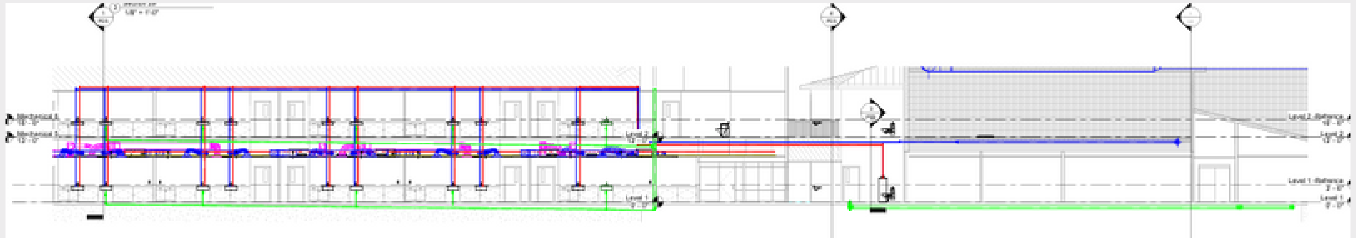

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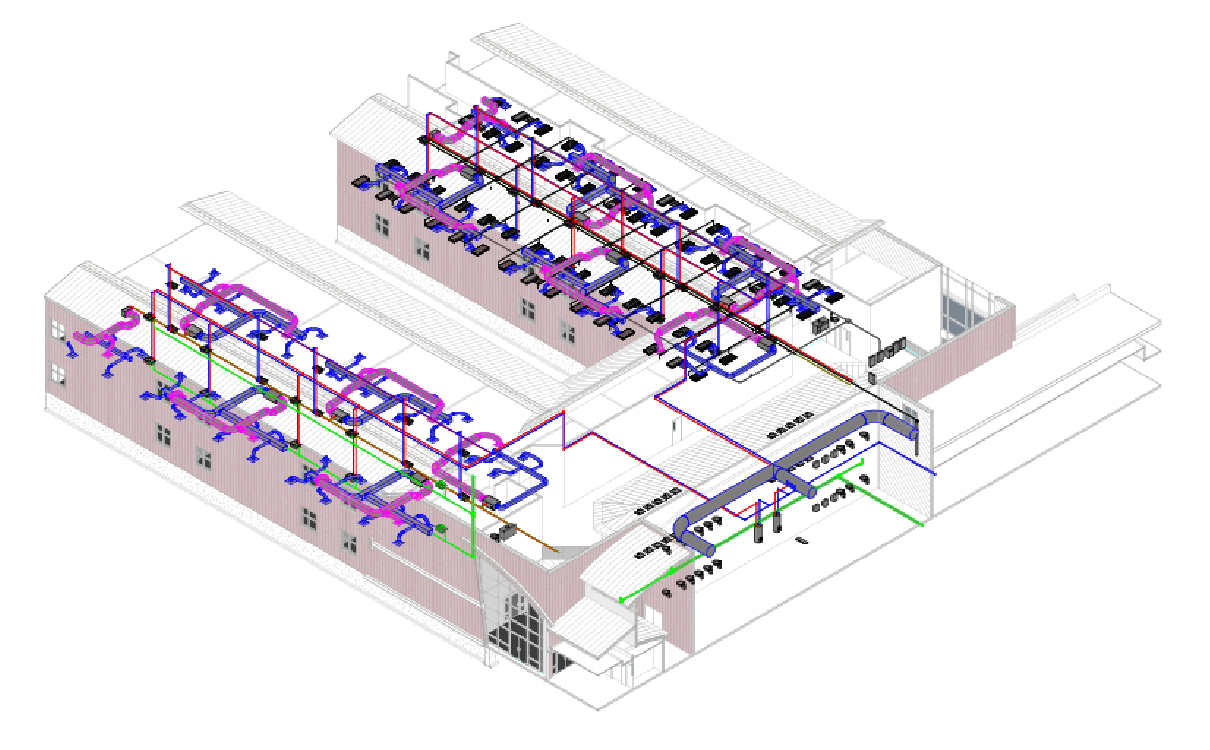
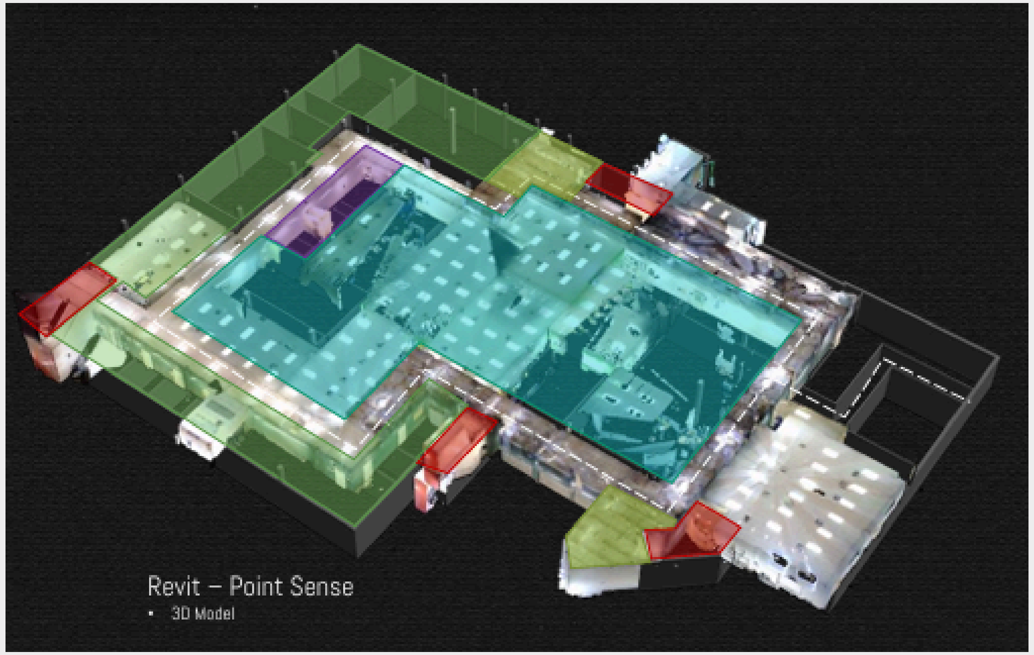


My systematic approach ensures clash-free installations that integrate seamlessly with building designs while maintaining full regulatory compliance.

EXISTING CONDITIONS

3D Modeling



VDC Methodology: Implemented an end-to-end digital workflow using 3D laser scanning (Faro Focus) to create an accurate digital twin of existing conditions. This coordinated Revit model served as the single source of truth for all design teams.

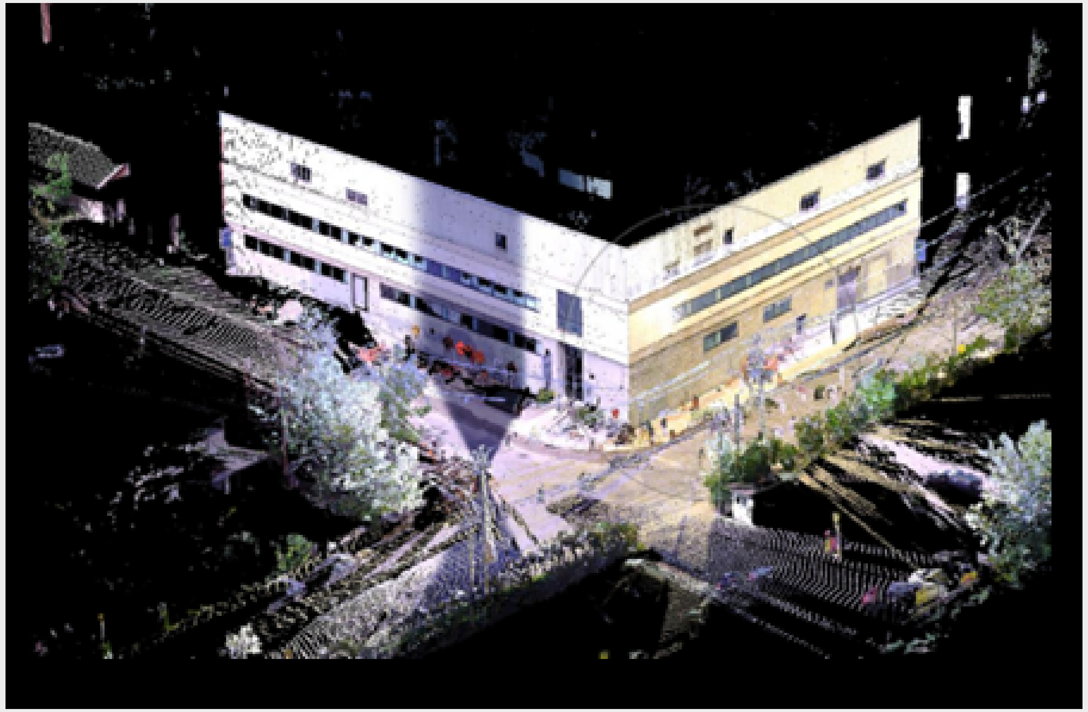
COMPREHENSIVE VDC PROJECT

Project Scope: Led a full-scope Virtual Design and Construction (VDC) project integrating architectural, structural, and MEP disciplines for the George Brown College campus expansion.

FARO
- 3D laser scanner

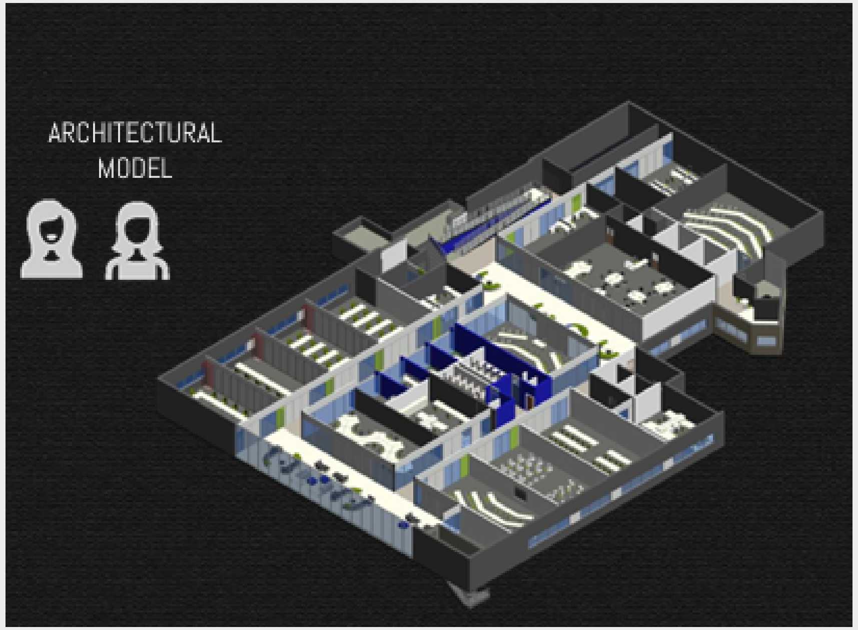
Scene
- Register the point cloud
- Procces the scans

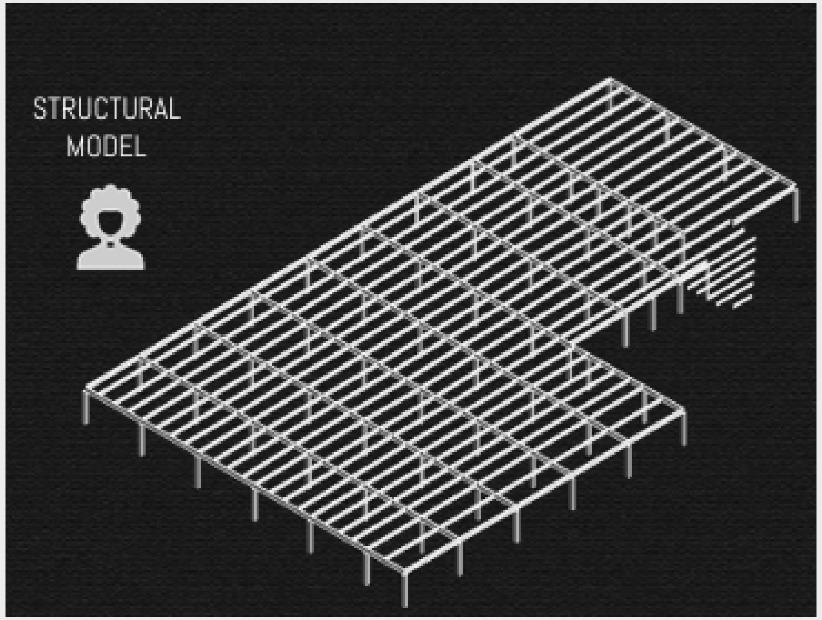
Autodesk Recap
- Clean the point cloud
- To have access to the 360 images

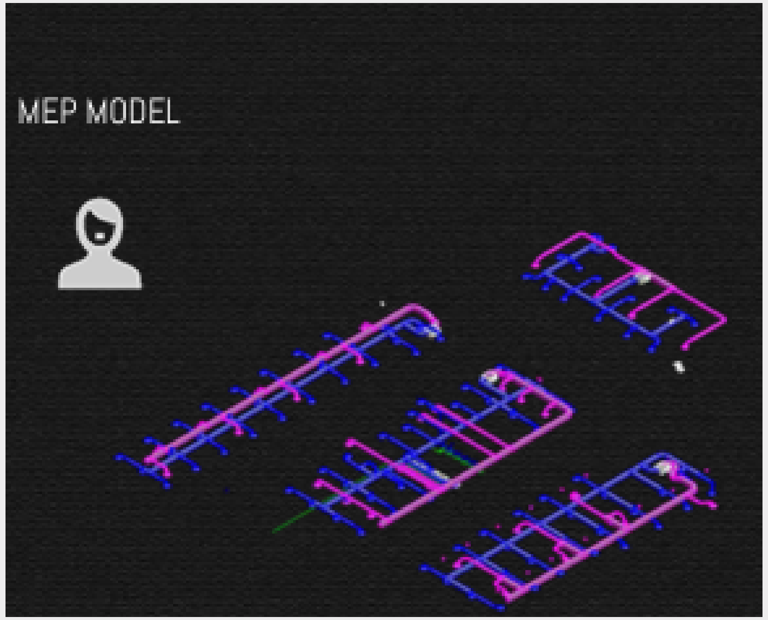


KEY DELIVERABLES

- Architectural: Existing conditions modeling and new interior layouts
- Structural: Structural analysis and reinforcement design
- MEP Systems: Complete mechanical (VRF systems) and electrical (power, lighting, data) design
- Coordination: Multi-disciplinary clash detection and resolution using Navisworks





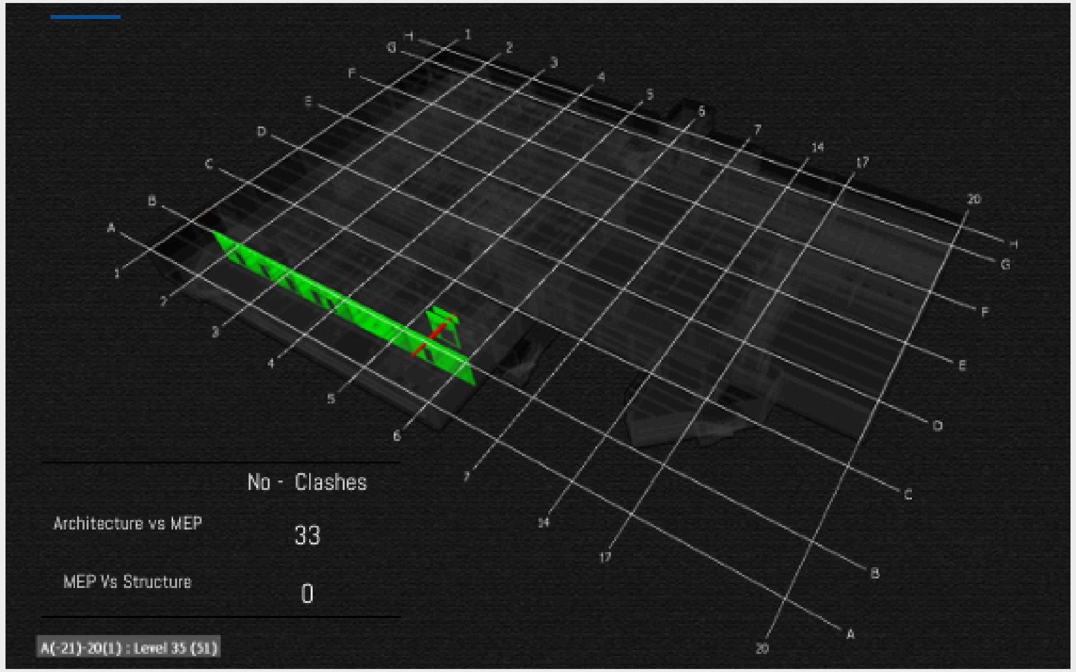


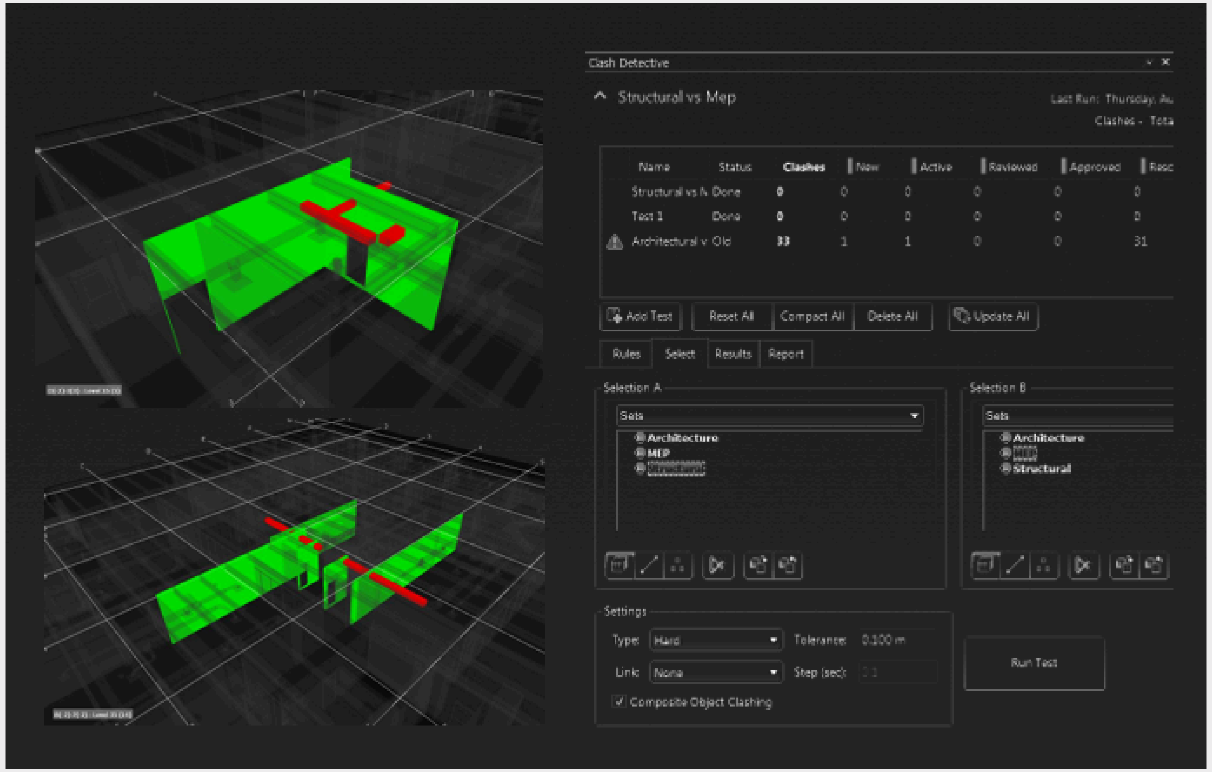
Outcome: Delivered construction-ready models and documentation, ensuring spatial coordination and reducing field conflicts. This project demonstrates comprehensive VDC capabilities across all building disciplines, from reality capture to constructible models.

CLASH DETECTION

Navisworks

- Structural vs. Architectural: Assigned to Structural and Architectural teams for resolution of frame/partition conflicts.
- MEP vs. Structural: Assigned to MEP team for ductwork, piping, and conduit conflicts with beams and openings.
- MEP vs. Architectural: Assigned to MEP and Architectural teams for ceiling, lighting, and fixture coordination.
- MEP Internal: Assigned to Mechanical and Electrical leads for internal system coordination.





Outcome:

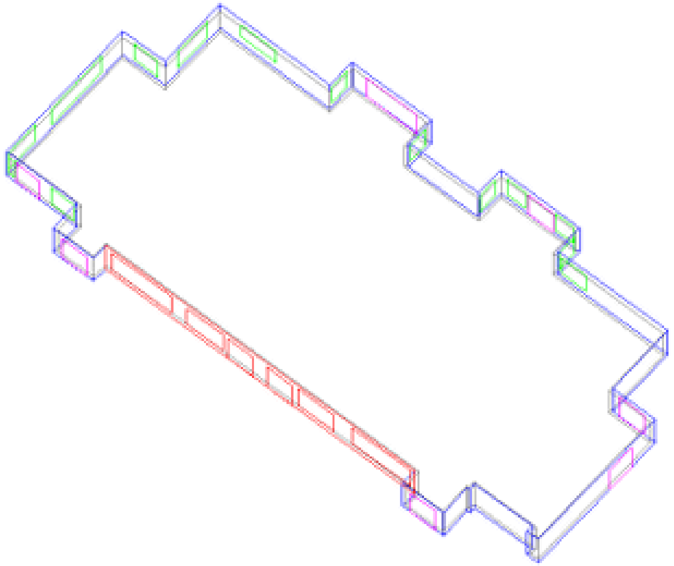
Resolved 50+ major clashes pre-construction, reducing on-site rework by ~30%.

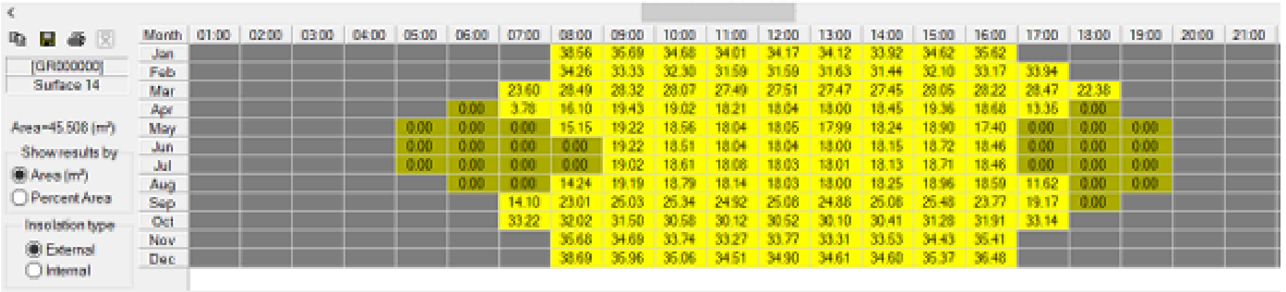
Streamlined communication through ACC's centralized platform, cutting resolution time by 25%.

Delivered a clash-free, constructible model, ensuring project adherence to schedule and budget.

For this scenario I picked the highest and the lowest wall for the three Levels:

Ground Floor Wall nº 1:





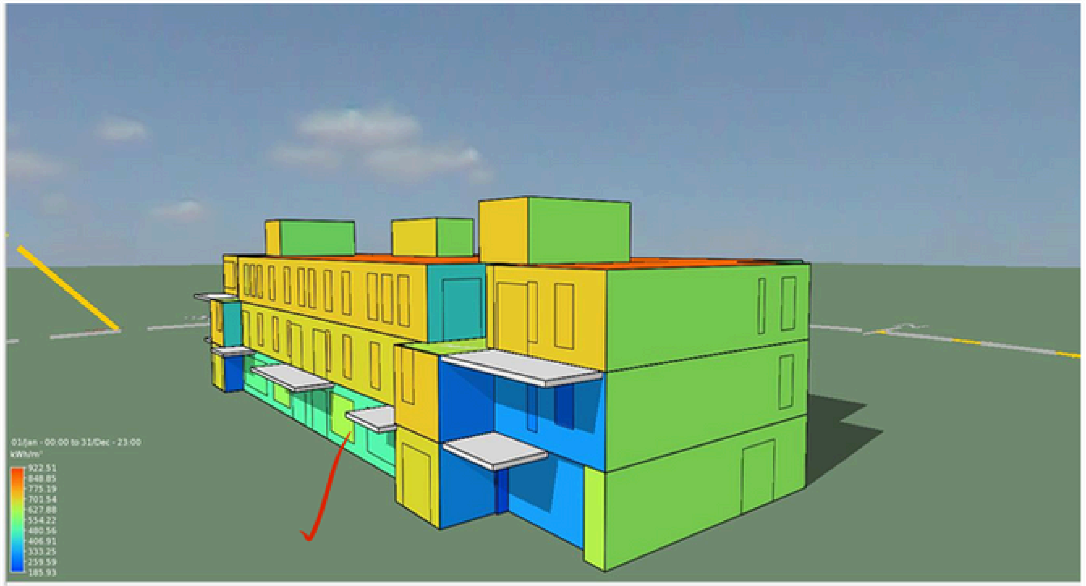
Month	01:00	02:00	03:00	04:00	05:00	06:00	07:00	08:00	09:00	10:00	11:00	12:00	13:00	14:00	15:00	16:00	17:00	18:00	19:00	20:00	21:00
Jan								38.56	35.69	34.68	34.01	34.17	34.12	33.92	34.62	35.62					
Feb								34.26	33.33	32.30	31.59	31.59	31.63	31.44	32.10	33.17	33.94				
Mar							23.60	28.49	28.32	28.07	27.49	27.51	27.47	27.45	28.05	28.22	28.47	22.38			
Apr						0.00	3.78	16.10	19.43	19.02	18.21	18.04	18.00	18.45	19.36	18.68	13.35	0.00			
May					0.00	0.00	0.00	15.15	19.22	18.56	18.04	18.05	17.99	18.24	18.90	17.40	0.00	0.00	0.00		
Jun					0.00	0.00	0.00	0.00	19.22	18.51	18.04	18.04	18.00	18.15	18.72	18.46	0.00	0.00	0.00		
Jul					0.00	0.00	0.00	0.00	19.02	18.61	18.08	18.03	18.01	18.13	18.71	18.46	0.00	0.00	0.00		
Aug						0.00	0.00	14.24	19.19	18.79	18.14	18.03	18.00	18.25	18.96	18.59	11.62	0.00	0.00		
Sep							14.10	23.01	25.03	25.34	24.92	25.08	24.88	25.08	25.48	23.77	19.17	0.00			
Oct							33.22	32.02	31.50	30.58	30.12	30.52	30.10	30.41	31.28	31.91	33.14				
Nov								35.68	34.69	33.74	33.27	33.77	33.31	33.53	34.43	35.41					
Dec								38.69	35.96	35.06	34.51	34.90	34.61	34.60	35.37	36.48					

Month	5:00	6:00	7:00	8:00	9:00	10:00	11:00	12:00	13:00	14:00	15:00	16:00	17:00	18:00	19:00	
Jan				38.56	35.69	34.68	34.01	34.17	34.12	33.92	34.62	35.62				
Feb				34.26	33.33	32.3	31.59	31.59	31.63	31.44	32.1	33.17	33.94			
Mar			23.6	28.49	28.32	28.07	27.49	27.51	27.47	27.45	28.05	28.22	28.47	22.38		
Apr		0	3.78	16.1	19.43	19.02	18.21	18.04	18	18.45	19.36	18.68	13.35	0		
May	0	0	0	15.15	19.22	18.56	18.04	18.05	17.99	18.24	18.9	17.4	0	0	0	
Jun	0	0	0	0	19.22	18.51	18.04	18.04	18	18.15	18.72	18.46	0	0	0	
Jul	0	0	0	0	19.02	18.61	18.08	18.03	18.01	18.13	18.71	18.46	0	0	0	
Aug		0	0	14.24	19.19	18.79	18.14	18.03	18	18.25	18.96	18.59	11.62	0	0	
Sep			14.1	23.01	25.03	25.34	24.92	25.08	24.88	25.08	25.48	23.77	19.17	0		
Oct			33.22	32.02	31.5	30.58	30.12	30.52	30.1	30.41	31.28	31.91	33.14			
Nov				35.68	34.69	33.74	33.27	33.77	33.31	33.53	34.43	35.41				
Dec				38.69	35.96	35.06	34.51	34.9	34.61	34.6	35.37	36.48				Total
	0	0	74.7	276.2	320.6	313.26	306.42	307.73	306.12	307.65	315.98	316.17	139.69	22.38		3006.9

IES ENERGY ANALYSIS REPORT

- Achieved 25% higher energy performance than OBC SB-10 standards through detailed IES energy modeling
- Reduced annual energy consumption by 35% via optimized HVAC scheduling and load management
- Enhanced building envelope performance by 23% using advanced insulation and glazing strategies
- Lowered cooling loads by 28% through passive design and solar shading analysis
- Enabled 40% solar integration potential with rooftop PV system feasibility studies

Delivered data-driven design solutions that exceeded sustainability targets while maintaining project viability.



South-East Elevation

Maryam Shojaei

GET IN TOUCH

Email:

Maryam.Shojaei@Georgebrown.ca

Location:

Toronto, ON

lwww.linkedin.com/in/maryam-shojaei5525